Science · BUILD · Autumn 1 · Week 8 · SCI-BUILD-A1-W08-P2 · Date _____

Use: Show what you know about body science | BUILD Science

Today I can… ☐ Show one idea about bones. Show how we move. Show what food does and where it comes from. ☐ Use because to join two science ideas. ☐ Use a model or the evidence. Show how it backs up what you say.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

STAFF ANSWERS · Supported

Arrival

1. The skull keeps the brain safe.
2. It starts to contract. It gets short and pulls.
3. The dog has a backbone.
4. J has less sugar per 100 g.

Exit

1. The biceps contracts. It gets short and pulls. Its partner is the triceps. It relaxes.
2. Food gives energy. We need it to move. Muscles use it. Food moves from grass to rabbit.

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STAFF ANSWERS · Standard

Arrival

1. The skull keeps the brain safe.
2. It starts to contract. It gets short and pulls.
3. The dog has a backbone.
4. J has less sugar per 100 g.

Exit

1. The biceps contracts. It gets short and pulls the forearm at the elbow. The triceps relaxes. Bones and muscles work as a team.
2. Food gives us nutrients. These parts of food give us energy. They help our body work. Muscles use it to contract. The arrow shows where food and energy go. They move from grass to rabbit.

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STAFF ANSWERS · Stretch

Arrival

1. The skull keeps the brain safe.
2. It starts to contract. It gets short and pulls.
3. The dog has a backbone.
4. J has less sugar per 100 g.

Exit

1. Muscles contract. They get short and pull bones at joints. The biceps contracts. It pulls to bend the arm. The triceps contracts. It pulls to make the arm straight. Its partner relaxes.
2. Food gives energy and parts to help us grow. Muscles use energy to contract. They pull bones at joints. The model leaves out tendons and other muscles. It leaves out the 3D shape and fine control.

Use: Show what you know about body science | BUILD Science

Today I can… ☐ Show one idea about bones. Show how we move. Show what food does and where it comes from. ☐ Use because to join two science ideas. ☐ Use a model or the evidence. Show how it backs up what you say.

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STAFF ANSWERS · reasoning

STAFF ANSWERS · BUILD Science

Shared activities

Bone jobs: support, protection and movement. Dog vertebrate, earthworm invertebrate. Biceps pulls to bend, triceps contracts to straighten. Grass → rabbit → fox. Wider feet kept the matched model upright more often in the supplied example. Sugar is one piece of nutrition evidence.

Supported task

Skull protects brain. Dog has the internal backbone.

Biceps pulls; triceps relaxes.

Carbohydrate provides energy. Wide feet: upright 5/5 times compared with 2/5 for narrow feet.

Grass → rabbit → fox. Arrows go from food to eater.

Food provides nutrients used for energy. Muscles use energy when they contract and pull bones.

Independent task

Skull protects brain or ribcage protects heart/lungs. Dog vertebrate; earthworm invertebrate.

Bending: biceps contracts, triceps relaxes. Straightening: triceps contracts, biceps relaxes.

J has 5 g sugar per 100 g, less than K at 10 g. Wider feet kept this matched model upright more often: 5/5 versus 2/5.

Grass → rabbit → fox. For example, food and energy pass from grass to rabbit. Grass makes its own food using light.

For example, food provides energy, which muscles use when they contract and pull bones. Protein helps growth/repair; calcium supports bones are other valid linked explanations.

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STAFF ANSWERS · reasoning · continued

Aspire task

Skull/ribs protect organs; muscles pull bones at joints. Earthworms have no backbone but can move, so that claim is false.

Triceps pulls to straighten the elbow as biceps relaxes. Biceps pulls to bend it. Muscles actively pull, not push.

J has half K's sugar per 100 g: 5 versus 10 g. Wide feet were upright 5/5, narrow 2/5. Five trials and imperfectly matched taps limit evidence; model results alone do not establish all human balance behaviour.

Grass → rabbit → fox. Food and energy transfer from eaten organism to eater. Grass makes food using light; the animals obtain it by eating.

Nutrients provide energy and materials; muscles use energy to contract and pull bones at joints. The model simplifies tendons, three-dimensional anatomy, other muscles and detailed control.

Supported

The biceps contracts. It gets short and pulls. Its partner is the triceps. It relaxes.

Food gives energy. We need it to move. Muscles use it. Food moves from grass to rabbit.

Independent

The biceps contracts. It gets short and pulls the forearm at the elbow. The triceps relaxes. Bones and muscles work as a team.

Food gives us nutrients. These parts of food give us energy. They help our body work. Muscles use it to contract. The arrow shows where food and energy go. They move from grass to rabbit.

Aspire

Muscles contract. They get short and pull bones at joints. The biceps contracts. It pulls to bend the arm. The triceps contracts. It pulls to make the arm straight. Its partner relaxes.

Food gives energy and parts to help us grow. Muscles use energy to contract. They pull bones at joints. The model leaves out tendons and other muscles. It leaves out the 3D shape and fine control.

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STAFF ANSWERS · checks

Yes. The stated backbone evidence supports the classification.

That does not follow. Animals without backbones can move in other ways.